# LAB REPORT: LAB 6 - API GATEWAY PATTERN AND ACCESS CONTROL

Kyte

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## 1. Cover Page

Title: API Gateway Pattern and Access Control for the Kyte Project Management Tool

Course Info: Software Architecture

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Date: Dec 16, 2025

## 2. Abstract/Summary


This lab describes how Kyte applies API gateway responsibilities in a single server entry point, centralizing routing and authentication before requests reach business logic.

## 3. Lab Specific Section: VI. API Gateway and Security

*Note: Kyte is a modular monolith. The gateway role is implemented inside* `packages/server/src/index.ts` *rather than as a separate service.*

### 3.1 Objectives

- Centralize routing into one entry point.
- Enforce authentication before requests reach domain services.
- Simplify client integration with a single base URL.

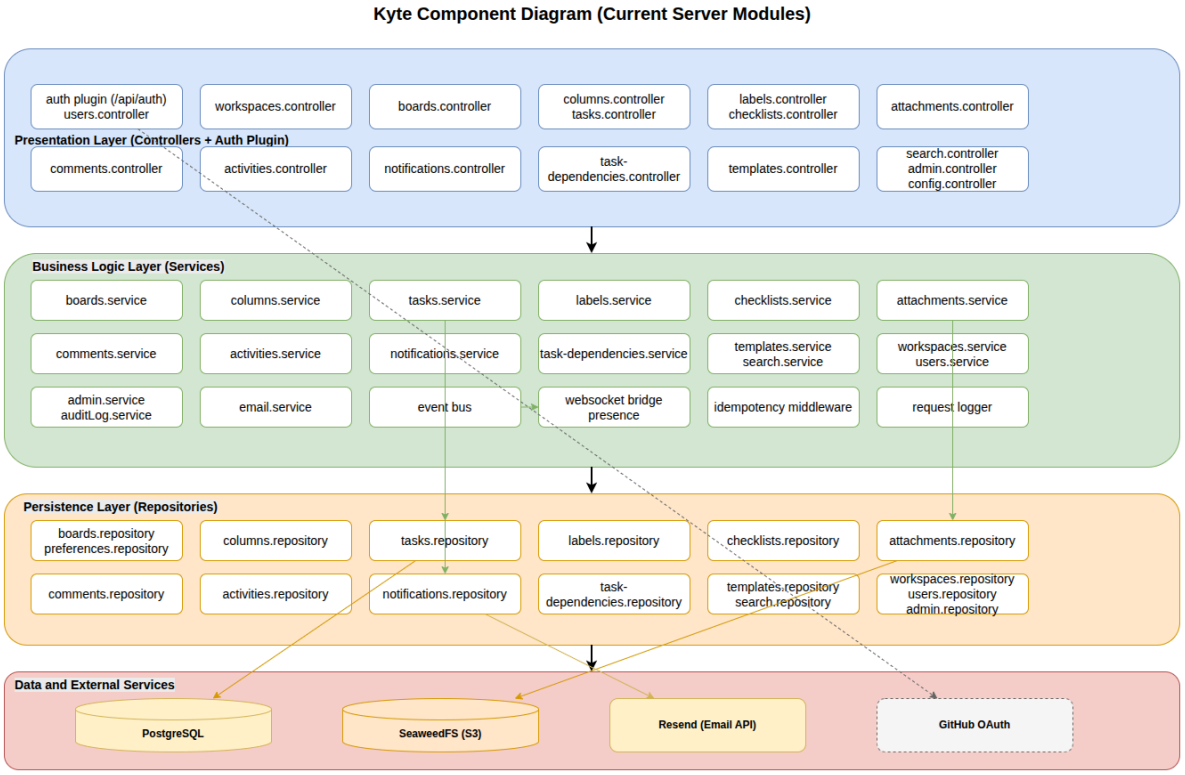
### 3.2 Technology and Tools

| Tool | Purpose |
|---|---|
| **Elysia** | Core framework for routing and middleware. |
| **Better Auth** | Authentication middleware for sessions and sessions. |
| **Eden Treaty** | Type safe API integration for the web client. |

### 3.3 Single Entry Point

The server entry point mounts all feature modules under a common API prefix:

```
const app = new Elysia()
  .use(cors())
  .use(betterAuthPlugin)
  .group('/v1', (group) =>
    group
      .use(authController)
      .use(boardsController)
      .use(tasksController)
      .use(workspacesController)
  )
  .listen(3000)
```




### 3.4 Authentication Middleware

- Incoming requests are validated by the auth middleware.
- If the session is invalid, the gateway returns `401` immediately.
- If valid, user context is attached and the request continues.

### 3.5 Cross Cutting Concerns

The gateway is the right place for consistent behavior across modules:

- Error formatting and status codes
- CORS policy enforcement
- Request logging for diagnostics

## 4. Conclusion & Reflection

Centralizing the gateway responsibilities improves security and simplifies client usage. It also provides a consistent place to enforce policies and add observability without duplicating code across modules.